FRI Natural language processing course 2026
Initial implementation report

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Natural language processing - Submission 2: Initial implementation and baseline results

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Abstract

General-purpose large language models often lack reliable knowledge of specific legal systems and may hallucinate unsupported legal claims. In this project, we build an initial retrieval-augmented generation baseline for Slovenian legal question answering. The system uses a subset of Slovenian legal documents from the PISRS dataset, splits legal texts into article-level chunks, embeds them with a multilingual retrieval model, stores them in a FAISS index, and generates answers with GaMS-2B-Instruct. Our current baseline focuses on real-estate and property-law-related legislation. The implemented system retrieves relevant legal sources and generates short, source-grounded answers (in Slovenian). On an initial evaluation set of 15 questions, the retriever achieved 88.89% top-1 accuracy and 100% top-3 accuracy on questions with manually specified expected sources/articles. We also analyse current limitations, including source contamination, noisy chunks, and too long generated answers, then outline future improvements.

Introduction

General-purpose large language models are not reliable enough for legal question answering on their own, especially for smaller jurisdictions such as Slovenia. They may generate plausible but unsupported legal claims, and they often lack direct access to Slovenian legislation. Our project therefore focuses on a retrieval-augmented generation (RAG) approach for Slovenian legal question answering. The goal is to answer user questions by first retrieving relevant legal provisions and then generating a short answer grounded in those sources.

In Submission 1, we proposed a Slovenian legal assistant based on the COLESLAW/PISRS legal corpus. In this submission, we present the first working baseline implementation and initial results. The current system focuses on a real-estate and property-law-related subset of Slovenian legislation.

Current implementation

We implemented a RAG pipeline using the COLESLAW 1.0 corpus, specifically the PISRS file ”register-predpisov.jsonl”. We selected this file because it contains the main register of Slovenian regulations and laws. For the first baseline, we filtered the corpus to a real-estate/property-law subset using keywords such as *nepremičnine*, *zemljišče*, *lastnina*, *kataster*, *hipoteka*, *služnost*, *zemljiška knjiga*, and *urejanje prostora*. The documents were split mostly into article-level chunks. The resulting dataset contains 23,592 legal chunks.

The implemented pipeline is:

```
PISRS legal text
 -> article-level chunks
 -> BGE-M3 embeddings
 -> FAISS vector index
 -> top-k retrieved legal sources
 -> GaMS-2B-Instruct answer generation
```

For retrieval, we use `BAAI/bge-m3`, a multilingual embedding model. It converts each legal chunk and each user question into a dense vector. These vectors are indexed using FAISS, which allows efficient similarity search. For generation, we use `cjvt/GaMS-2B-Instruct`, a Slovenian-focused instruction-tuned model. Importantly, GaMS does not receive embeddings directly; it receives the raw retrieved legal text and generates an answer from that context.

The system runs on the ARNES Slurm cluster. We use the shared PyTorch Apptainer container:

```
/d/hpc/singularity/pytorch-24.12-py3.sif
```

Project files, the FAISS index, and the shared Hugging Face cache are stored in:

```
/d/hpc/projects/onj_fri/pad-siew/law-rag
```

Each team member uses a private Python virtual environment in their home directory. We verified that GaMS-2B-Instruct runs on a Tesla V100S 32GB GPU, and that BGE-M3 embeddings and FAISS retrieval work correctly.

Initial results and analysis

We evaluated the baseline on 15 manually written Slovenian legal questions from the real-estate/property-law domain. The questions cover concepts such as pogodbena komasacija, služnost, etažna lastnina, hipoteka, zemljiška knjiga, posest, lastninska pravica, gradbeno dovoljenje, and urejanje prostora. For 9 of these questions, we manually specified an expected source or article, which allowed us to compute retrieval accuracy.

The retrieval results were:

Metric	Result
Top-1 retrieval accuracy	8/9 = 88.89%
Top-3 retrieval accuracy	9/9 = 100.00%
Top-5 retrieval accuracy	9/9 = 100.00%

Top-1 accuracy means that the expected legal source or article was ranked first by the retriever. Top-3 accuracy means that the expected source appeared anywhere among the first three retrieved chunks. This is important for RAG because the generator usually receives more than one retrieved chunk. In our case, the correct source appeared in the top three results for every labelled question, which shows that the BGE-M3 + FAISS retrieval baseline works well for this restricted domain.

Several examples worked particularly well. For the question *"Kako se vpiše pogodbena komasacija?"*, the top retrieved source was *Zakon o urejanju prostora (ZUreP-3), 170. člen*. The generated answer correctly stated that contractual land consolidation is registered according to cadastral regulations and that the registration procedure is handled with priority. This answer was short, directly supported by the retrieved source, and did not include unrelated information.

For the question *"Kaj pomeni služnost?"*, the top retrieved source was *Stvarnopravni zakonik (SPZ), 210. člen*. The generated answer correctly described servitude as the right to use another person's thing or require the owner to refrain from certain actions. This is a good example of the system working as intended: the retriever found the exact legal definition, and GaMS produced a concise answer based on that article.

Other strong examples include *"Kaj pomeni posest?"*, where the system retrieved *SPZ, 24. člen* and answered that possession is direct factual control over a thing, and *"Kaj je lastninska pravica?"*, where it retrieved *SPZ, 37. člen* and generated a correct description of ownership as the right to possess, use, enjoy, and dispose of a thing.

The evaluation also showed that the system can still answer correctly even when the best source is not ranked first. For example, for *"Kaj je gradbeno dovoljenje?"*, the first retrieved result came from *Zakon o varstvu podzemnih jam*, which is not the best general source for this question. However, relevant articles from *Gradbeni zakon (GZ-1)* appeared in ranks 2 and 3. This means that top-3 retrieval was good enough to provide the generator with useful context, but it also shows that the top-1 ranking still needs improvement.

The main error type we observed is source contamination. For the question *"Kaj je hipoteka?"*, the retriever returned two relevant articles from *Stvarnopravni zakonik*, but it also retrieved an article from *Pomorski zakonik* about ship mortgages. The generated answer partly included this maritime-law information. This is not a hallucination in the strict sense, because the model used retrieved text, but it is still an error because the source was not relevant to the intended legal context. For legal QA, we therefore treat irrelevant but source-derived information as a form of contamination.

A second issue is verbosity and incomplete answers. For broad questions such as *"Kaj pomeni urejanje prostora?"*, the retrieved article contains a long legal definition with many goals and principles. The model began producing a long answer and the response was eventually cut off by the generation limit. This suggests that prompt instructions such as "answer in at most two sentences" are useful, but not always sufficient. We may also need to reduce the amount of context passed to the model or make the prompt more extractive.

A third issue is ambiguity in some questions. For example, questions about *kataster nepremičnin* or *zemljiški kataster* can retrieve related but not always ideal sources. In one case, the system retrieved articles from *Stanovanjski zakon* or *Zakon o zemljiški knjigi*, even though a more specific cadastral regulation might be more appropriate. This may be caused by the restricted real-estate subset, keyword-based filtering, or the fact that similar terms appear across several laws. It shows that expanding and cleaning the corpus may improve coverage.

Overall, the initial results are encouraging. The retrieval component performs well on the manually labelled examples, and GaMS-2B-Instruct is able to generate understandable Slovenian answers grounded in retrieved legal text. The most important remaining problems are not basic infrastructure problems, but quality issues: selecting the most relevant sources, avoiding unrelated retrieved context, cleaning noisy chunks, and controlling answer length.

Future ideas

- Improve article chunking and remove trailing headings or unrelated text from chunks.

- Add more manually labelled evaluation questions and evaluate answer correctness, groundedness, source quality, and hallucinations.

- Improve prompt instructions so answers are shorter, more extractive, and use only directly relevant sources.
- Experiment with passing fewer retrieved chunks to the generator to reduce source contamination.
- Add hybrid retrieval, combining dense BGE-M3 retrieval with keyword/BM25 search.
- Add a reranking step before generation to filter irrelevant sources.
- Compare GaMS-2B-Instruct with larger models such as GaMS-9B-Instruct if time and GPU resources allow.
- Consider PEFT/LoRA fine-tuning later, after building a reliable legal QA training/evaluation set.

Conclusion

We implemented a first working baseline for Slovenian legal question answering using retrieval-augmented generation. The system uses a real-estate-related subset of PISRS legislation from COLESLAW, article-level chunks, BGE-M3 embeddings, FAISS retrieval, and GaMS-2B-Instruct generation. The baseline runs on the ARNES Slurm cluster and produces source-grounded Slovenian answers. The initial results are good. On manually labelled questions, the correct source/article appears in the top three retrieved chunks for all labelled examples, and the generated answers are often concise and legally grounded. At the same time, the system still has limitations, especially source contamination, noisy chunks, and overlong answers. These issues provide clear directions for the next development stage.